Jianfeng Lin

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Education

- **Georgia Institute of Technology (GT), Atlanta, GA** *Aug. 2023 - Present*
  PhD in Quantitative Biosciences (QBioS)
  Advisor: Daniel I. Goldman
- **Wuhan University (WHU), Wuhan, China** *Sept. 2019 - June 2023*
  B.Eng in Mechanical Engineering
  GPA: 3.64/4.0

Experience

- **Complex Rheology and Biomechanics Lab at Georgia Tech** *May 2024 - Present*
  Graduate Research Assistant
  Advisor: Daniel I. Goldman
- **School of Physics, Georgia Tech** *Aug. 2023 - May 2025*
  Graduate Teaching Assistant
  PHYS 2211 - Intro Physics I: Fall 2023, Summer 2024, Spring 2025, Summer 2026
- **Dynamics Locomotion Group at Max Planck Institute for Intelligent Systems** *July 2022 - Sept. 2023*
  Research Assistant Remote
  Advisor: Alexander Badri-Spröwitz
- **Exoskeleton Lab at Wuhan university** *Aug. 2020 - June 2023*
  Undergraduate Research Assistant
  Advisor: Zhao Guo

Patents and Publications

Selected Papers, S=In Submission, J=Journal, C=Conference, A=Abstracts, P=Patent

† Equally contributed

In Submission

[S.3] Z. J. Xu, J. He, D. Aydan, M. Taylor, T. Wang, **J. Lin**, W. Dyer, D. I. Goldman. A Robust Antenna Provides Tactile Feedback in a Multi-legged Robot.

[S.2] B. Chong†, T. Wang†, K. Diaz, C. J. Pierce, E. Erickson, J. Whitman, Y. Deng, E. Flores, R. Fu, J. He, **J. Lin**, H. Lu, G. Sartoretti, H. Choset, D. I. Goldman. The Omega Turn: A General Turning Template for Elongate Robots.

[S.1] B. Chong, J. He, D. Irvine, T. Wang, E. Flores, D. Soto, **J. Lin**, Z. Xu, V. R. Nienhusser, G. Blekherman, D. I. Goldman. Robust control for multi-legged elongate robots in noisy environments.

Journal Papers

[J.3] **J. Lin**†, T. Wang†, B. Chong, M. Fernandez, Z. Xu, D. I. Goldman. Optimal swimming with body compliance in an overdamped medium. *Physical Review Applied*, 2026, DOI: 10.1103/pbkh-py5h

[J.2] J. He, B. Chong, **J. Lin**, Z. Xu, H. Bagheri, E. Flores, D. I. Goldman. Probabilistic approach to feedback control enhances multi-legged locomotion on rugged landscapes. *IEEE Transactions on Robotics*, 2025, DOI: 110.1109/TRO.2025.3593133

[J.1] **J. Lin**, J. Ke, R. Xiao, X. Jiang, M. Li, X. Xiao, Z. Guo. Bioinspired bidirectional stiffening soft actuators enable versatile and robust grasping. *Soft Robotics*, 2024, DOI: 10.1089/soro.2022.0212

Conference Papers

[C.3] M. Iaschi, B. Chong, T. Wang, **J. Lin**, J. He, D. Soto, Z. Xu, D. I. Goldman. Addition of a peristaltic wave improves multi-legged locomotion performance on complex terrains. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, DOI:10.1109/ICRA55743.2025.11127656

[C.2] **J. Lin**, Z. Guo, A. Badri-Spröwitz. **Bird-inspired tendon coupling improves paddling efficiency by shortening phase transition times**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, DOI: 10.1109/ICRA55743.2025.11128691

[C.1] **J. Lin**, R. Xiao, Z. Guo. Design and Stiffness Analysis of a Bio-Inspired Soft Actuator with Bi-Direction Tunable Stiffness Property. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, DOI: 10.1109/IROS55552.2023.10341730

Selected Conference Abstracts/Posters

[A.1] **J. Lin**, Z. Guo, A. Badri-Spröwitz. Bird-inspired tendon coupling improves paddling efficiency by shortening phase transition times. *ICRA 2025 Workshop "Towards Agility and Robustness: Mechanical Intelligence in Robotics"*, 2025. **Best Design Award**

Patent

[P.1] Z. Guo, **J. Lin**, R. Xiao. A rigid-soft coupling actuator with multi-directional variable stiffness properties and a manufacturing method. *China Patent*, CN114952937A, 2022

Honors and Awards

- **Georgia Tech School of Physics Travel Grant** ($700) *Feb. 2026*
- **Georgia Tech College of Sciences Graduate Career Connect Travel Grant** ($500) *Dec. 2025*
- **QBioS Travel Award**, covering the expense to SICB 2026 ($1000) *Aug. 2025*
- **Best Design Award** in the ICRA 2025 Workshop, Mechanical Intelligence in Robotics ($250) *May 2025*
- **The Second Prize** in the 10th National College Mechanical Innovation Competition of China *Aug. 2022*

Academic Services

- **Organizer** of Robophysics Mini Boot Camp (NSF PoLS workshop) *2024, 2025*
- **Organizer** of Robophysics Mini Boot Camp (NSF PoLS workshop) *2024, 2025*
- **Reviewer**: Scientific Reports, IEEE RA-L *2025 - Present*

Professional Memberships

- **IEEE**, Student Member *2022 - Present*
- **Society of Comparative and Integrative Biology**, Member *2025 - Present*
- **American Physical Society**, Member *2025 - Present*

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